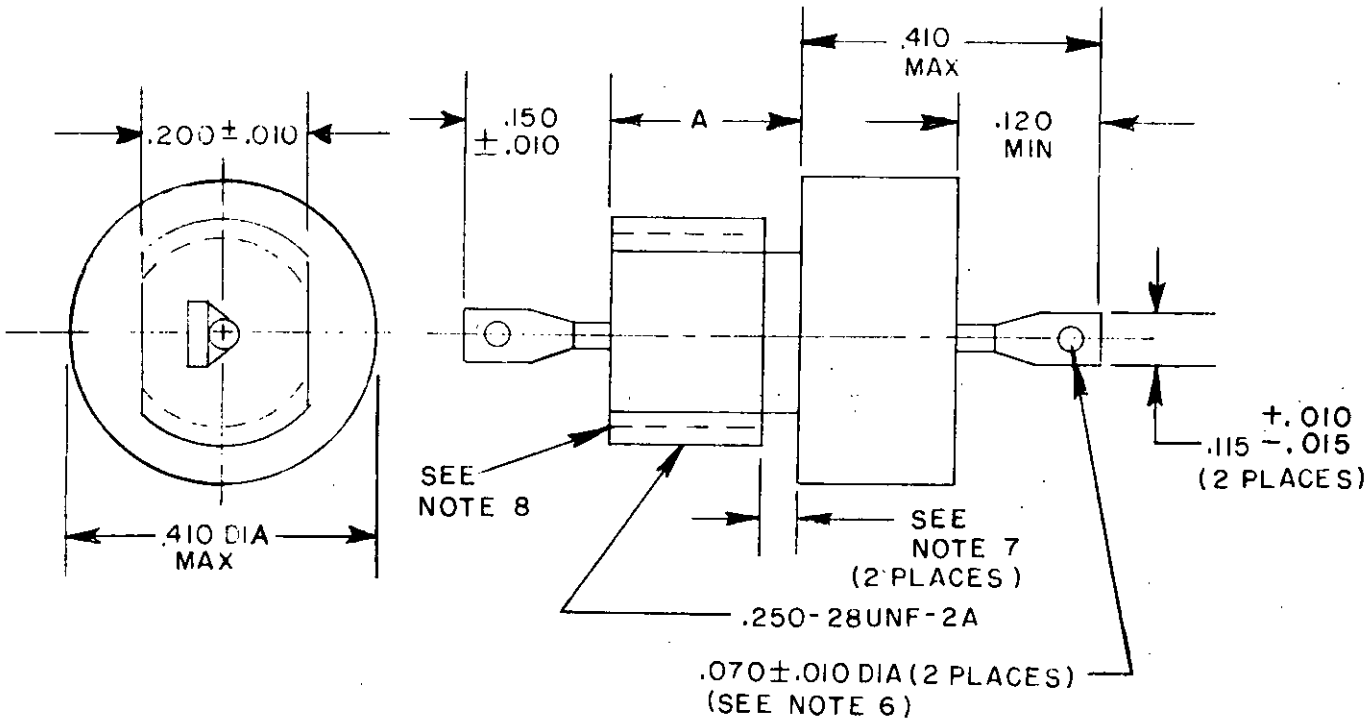
MILITARY SPECIFICATION SHEET

FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLES FS10, FS11, FS12, FS13, AND FS14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-28861.



Styles FS10, FS11, FS12, FS13 and FS14.

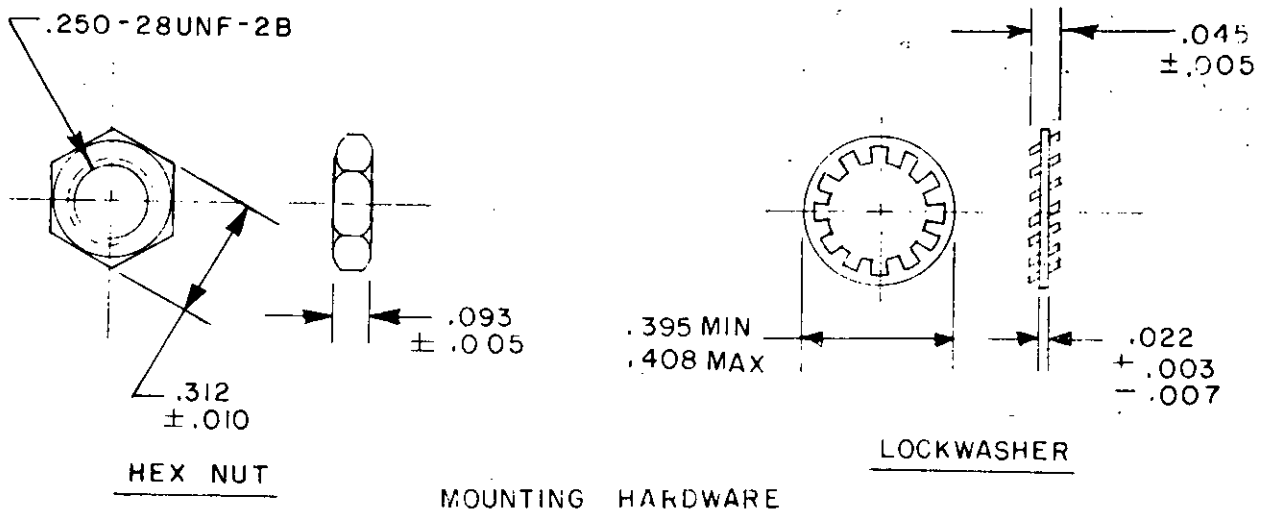
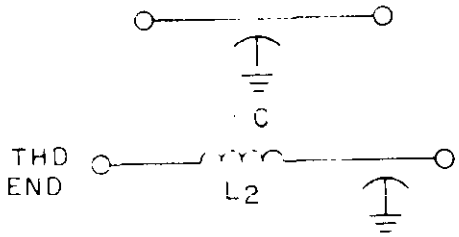


FIGURE 1. Case and hardware dimensions and circuit diagrams.

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FSC 5915 - 5910

Dash no.	A dimension (±010)
001 THRU 010	.187
011 THRU 020	.312



CIRCUIT DIAGRAMS

INCHES	MM	INCHES	MM
.003	.08	.115	2.92
.005	.13	.120	3.05
.007	.18	.150	3.81
.010	.25	.187	4.75
.015	.38	.200	5.08
.022	.56	.250	6.35
.035	.89	.312	7.92
.045	1.14	.395	10.03
.050	1.27	.400	10.16
.070	1.78	.408	10.36
.093	2.36	.410	10.41

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
5. Terminal identification (non-symmetrical filters). The case shall be marked at the threaded end of the filter with the symbol L or the circuit diagram shall be marked on the case.
6. Optional slot may be supplied, .050 ±.010 inches x .070 ±010 inches.
7. Imperfect thread or undercut optional .050 inches maximum.
8. One imperfect thread allowed .035 inches maximum.
9. Recommended mounting torque: 44 oz-in ±4 oz-in.

FIGURE 1. Case and hardware dimensions and circuit diagram - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 5 grams, maximum.

Case finish:

Class B filters, T only (tin-plated or tin-lead plated).

Class S filters, T or G only (tin-plated, tin-lead plated or gold plated). In accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55° to +125°C.

Rated voltage: As specified in table I.

Rated current: 15 amperes.

Capacitance: See table I for minimum capacitance value.

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohms, minimum.

At +125°C: 100 megohms, minimum.

Insertion loss: See table I.

Voltage drop: 0.12 Vdc, maximum.

DC resistance: 0.008 ohms, maximum.

Seal: In accordance with MIL-F-28861. Leakage rate for class S filters shall not exceed 1×10^{-7} atm cc/sec.

Temperature rise: 25°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Marking: Filters and capacitors shall be marked, as a minimum, with the following information:

a. Military part number.

b. JAN brand.

c. Source code.

d. Date code.

e. Terminal identification or circuit diagram (non-symmetrical filters only).

In addition, full marking, in accordance with MIL-F-28861 shall be marked on the package.

Product assurance level: In accordance with table I.

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TABLE I. Electrical characteristics.

Dash number	Product assurance level Class B	Product assurance level Class S	Style	Circuit	Rated voltage DC volts	Rated voltage AC 2/ volts	Minimum Capacitance (µF)	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/3/ At +25°C: 30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz	At -55° and +125°C: 30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz
001, 011	X		FS10	L2	50	-	1.2	15	28	33	40	40	70	70	13	26	31	38	38	70	70
002, 012	X		FS10	C	50	-	1.2	15	28	33	40	40	70	70	13	26	31	38	38	70	70
003, 013	X	X	FS11	L2	70	-	0.7	10	24	30	40	40	64	70	8	22	28	38	38	64	70
004, 014	X	X	FS11	C	70	-	0.7	10	24	30	40	40	64	70	8	22	28	38	38	64	70
005, 015	X	X	FS12	L2	100	-	0.45	6	19	25	36	40	60	70	4	17	23	34	38	60	70
006, 016	X	X	FS12	C	100	-	0.45	6	19	25	36	40	60	70	4	17	23	34	38	60	70
007, 017	X	X	FS13	L2	150	-	0.25	-	14	20	31	40	56	70	-	12	18	29	38	56	70
008, 018	X	X	FS13	C	150	-	0.25	-	14	20	31	40	56	70	-	12	18	29	38	56	70
009, 019	X	X	FS14	L2	200	125	0.15	-	10	16	26	40	52	70	-	9	14	25	38	52	70
010, 020	X	X	FS14	C	200	125	0.15	-	10	16	26	40	52	70	-	9	14	25	38	52	70

1/ Insertion loss measurements shall be made under full load over the frequency range of 150 kHz to 10 MHz. Insertion loss measurements above or below this frequency range shall be made under no load.

2/ 0 to 400 Hz.

3/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

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Part number: The part number shall be as follows:

M28861/01 -	XXX	X	X
	Dash no. per figure 1 and table I	Case finish	Product assurance level per MIL-F-28861 and table I.

Cataloging information. Dash numbers 002, 004, 006, 008, 010, 012, 014, 016, 018, and 020 shall be cataloged under FSC 5910 as feed through ceramic capacitors. Dash numbers 001, 003, 005, 007, 009, 011, 013, 015, 017, and 019 shall be cataloged under FSC 5915 as radio frequency interference filters.

Custodians:
Army - ER
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Army - AR, MI
Air Force - 17, 85, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - CG, MC, AS, OS, SH
Air Force - 19

Agent:
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Preparing activity:
Navy - EC

(Project 5915-0219-1)

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